

SCHOOL OF SCIENCE & TECHNOLOGY

FOR THE PLOT

Submitted by

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SWE4900A: SOFTWARE PROJECT II

DECLARATION

I declare that this is my original work and has never been submitted to any institution for the Certificate, Diploma, or Degree award. This document has been submitted to fulfill the requirements for a Degree at USIU University.

STUDENT: LEE BARAKA NGARI

Signature:…….

Date:…….. This project has been submitted for examination

purposes with the approval of the supervisor

SUPERVISOR: DR. FREDRICK M. OGORE

Signature:…… Date:……

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DEDICATION

To those of use who spent 150,000+ minutes on Spotify this year: this one’s for you. The ones with playlists for moods that don’t exist yet, who know the perfect song for every tiny moment, and whose "On Repeat" is basically a diary, may your recommendations never fail, your skip button gather dust, and your listening stay legendary. You set the vibe; the rest of us are just trying to catch up..

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The modern media landscape is defined by convenience and vast content variety. Platforms such as Netflix, Disney+, and Amazon Prime have transformed entertainment access, accelerating the dominance of on-demand streaming. This shift has normalized solitary viewing, replacing the traditional communal experience of cinemas or shared living rooms.

In parallel, digital socialization has become increasingly important, particularly during events such as the COVID-19 pandemic, which moved much of social life online. However, streaming platforms still lack integrated tools for live interaction, leaving users dependent on fragmented third-party solutions. Fortheplot addresses this gap by merging streaming with real-time social engagement, creating an experience that is both accessible and socially meaningful.

1.2 Background Information

The viability of this project rests on the convergence of strong economic and social forces. Economically, the global video streaming market was valued at USD 674.25 billion in 2024 and is projected to reach USD 2.66 trillion by 2032, with a compound annual growth rate of 18.5% (Fortune Business Insights, 2024; Grand View Research, 2024). This growth is fueled by the rise of Over-the-Top (OTT) platforms and subscription-based models, which consumers favor for convenience and content variety. The data confirms both the size and rapid expansion of the market, providing a strong foundation for innovative platforms that address unmet needs.

Socially, there is a pressing need to combat rising loneliness, especially among young adults. Between 2019 and 2021, 61% of young adults reported serious loneliness (American Psychological Association, 2022), and other studies show that one in three individuals aged 18–29 regularly feel isolated (Newport Institute, n.d.). While digital tools promise connection, research suggests they often fail to deliver meaningful interactions (Bonsaksen et al., 2021).

This creates a unique intersection: the booming demand for digital video content alongside a widespread crisis of social isolation (Deloitte, 2023). Evidence of consumer interest is already visible in the popularity of "watch party" and co-viewing formats on Twitch and YouTube, as well as Spotify Jam's success in collaborative listening, which reached 100 million monthly hours . The desire for shared media experiences is clear, yet current solutions are fragmented. The true gap lies not in demand, but in the absence of a dedicated, reliable platform built to merge streaming and social connection.

1.3 Problem Statement

Although numerous streaming platforms exist, a major market gap remains: the absence of a native, all-in-one solution for synchronized group viewing. Leading services such as Netflix, Disney+, and Amazon Prime are designed for individual, on-demand use, making shared viewing difficult. As a result, users rely on third-party tools that are fragmented and unreliable.

Browser extensions like Teleparty and Scener attempt to bridge this gap but face serious limitations. Users report frequent synchronization drift, where videos pause or jump unexpectedly, disrupting immersion. The experience is further fragmented, often requiring a mix of apps, streaming service, extension, and separate communication platforms like Zoom

or Discord. These tools are also technically fragile, breaking whenever streaming platforms update their interfaces. Mobile optimization is poor, with apps described as buggy or offering only partial functionality. In addition, moderation features such as host controls and private room settings are often missing, reducing usability for both casual and public group viewings.

User feedback highlights a deeper issue: current tools lack reliability and cohesion. Even when features are present, they often fail to work consistently. The opportunity for Fortheplot lies not simply in offering features but in delivering a seamless, integrated platform where playback, chat, and social interaction are designed to function together reliably.

1.4 Main Objective

To develop a native, all-in-one platform for synchronized movie watching, combining the convenience of streaming with the social dynamism of live interaction.

1.5 Specific Objectives

- To develop a robust movie streaming platform with user authentication, a comprehensive movie catalog, and HLS-based adaptive video playback.
- To implement a real-time room system supporting private and public rooms with integrated moderation and access controls.
- To achieve frame-accurate video synchronization across all participants to eliminate playback drift.
- To integrate a multi-modal communication suite, including in-room text chat and optional peer-to-peer video calling.

1.6 Justification

The development of Fortheplot is justified by both social and market needs. Socially, it addresses the growing issue of loneliness by enabling shared, synchronous viewing experiences that foster genuine connection. Unlike passive social media, which can heighten

isolation, a platform built for active, meaningful interaction can support digital well-being (Newport Institute, n.d.).

From a market standpoint, Fortheplot fills a critical gap left by unreliable third-party workarounds. By delivering a purpose-built, all-in-one solution, it offers a superior alternative aligned with the rapid growth of the video streaming industry.

Technically, the project is highly feasible. It leverages proven technologies, including React, Spring Boot, HLS.js, WebSockets, and WebRTC, to build a scalable and reliable application. Rather than inventing new tools, Fortheplot combines established technologies in an innovative way to meet the demands of synchronized streaming and real-time communication.

CHAPTER 2: LITERATURE REVIEW

2.1 Global Competitive Analysis

The market for synchronized group viewing is currently served by several third-party applications and browser extensions, but none provide a seamless, reliable solution. Their popularity demonstrates strong demand, yet their limitations create a clear opportunity for Fortheplot.

Teleparty (formerly Netflix Party) is the most widely used, with over 20 million users. It synchronizes playback and adds chat to services like Netflix and Disney+. However, it suffers from frequent desynchronization, playback errors, and a buggy mobile app often described as ineffective. As a browser extension, it is also fragile and prone to breaking whenever streaming services update their interfaces.

Scener offers synchronized viewing and integrated video chat for premium users. Its biggest drawback is that all participants must hold active subscriptions to the streaming platforms they want to use, creating a barrier to entry. Mobile features are also limited, allowing only partial participation without hosting or automatic synchronization.

Kast supports multiple sources, including local files, but users frequently report synchronization issues, reliability problems, and a confusing interface.

Taken together, these tools expose a fundamental weakness: they function only as wrappers around existing paid services, leaving the social layer fragile and secondary. Fortheplot's advantage lies in eliminating this dependency. By serving as both the content and social platform, it delivers a fully integrated experience where synchronized viewing and real-time interaction are the core product, not an add-on.

2.2 Market Dynamics in Kenya and Africa

Although Fortheplot is globally relevant, its success depends on adapting to regional market dynamics, particularly in Kenya. The African streaming market faces intense competition but is shaped by affordability challenges (Deloitte, 2023).

Studies from South Africa highlight a preference for local content and lower-priced services. Platforms like DStv and Showmax dominate, while Netflix lags due to higher costs (Fabricdata, 2025). Piracy, accounting for 13% of the South African market, further reflects financial barriers and the need for affordable solutions. These trends suggest that replicating Western business models will not succeed; instead, platforms must prioritize affordability and culturally relevant content.

Kenya's digital ecosystem is highly social, with WhatsApp, Facebook, and YouTube as leading platforms (Ngethe et al., 2021; ResearchGate, 2019). TikTok's rapid growth, along with startups like Chpter that enable social-first commerce, shows strong demand for tools that blend entertainment with community interaction (Tech Trends KE, 2025; Tech in Africa, n.d.).

For Fortheplot to succeed in Kenya, it should adopt a free or low-cost model while integrating local African storytelling into its content strategy. This combination of affordability and cultural relevance will be critical for long-term adoption and impact (Deloitte, 2023).

2.3 Merits and Demerits of Existing Systems

Despite their flaws, current group-viewing tools validate the core idea. Their popularity shows clear demand for synchronized playback and integrated chat, proving that users actively seek a shared viewing experience. Millions of installs confirm this unmet need.

However, their limitations expose the weaknesses of the "wrapper" model. The most significant is synchronization drift, where playback frequently falls out of sync, forcing users to manually adjust and disrupting immersion. Functional fragmentation is another issue, as

viewers often need three or more separate apps, streaming, extensions, and chat, to achieve one goal. These tools also lack robust host controls for managing users, private rooms, and permissions, limiting their usefulness for both private and public events. Finally, their technical fragility makes them unreliable, as even minor updates from streaming services can break functionality and require constant fixes.

2.4 The Unique Proposition of Fortheplot

Fortheplot is purpose-built to overcome the weaknesses of current solutions and deliver a superior, reliable service. Unlike fragile extensions layered onto existing platforms, it is designed as a native system for synchronized streaming.

The platform enhances service delivery through:

- **Seamless Integration**: Video playback, chat, and optional video calling are combined into one cohesive web application, removing the need for multiple apps and ensuring a fluid user experience.

- **Robust Synchronization**: A dedicated synchronization engine uses buffer management and real-time event handling to minimize drift, guaranteeing consistent playback across all users.

- **Device-Agnostic Design**: With a mobile-responsive, web-first approach, Fortheplot ensures a smooth experience across desktops and mobile devices, avoiding the poor optimization seen in competing tools.

CHAPTER 3: METHODOLOGY

3.1 Methodology

The project will follow an Agile Scrum methodology with two-week sprints. This iterative approach supports continuous development, testing, and feedback, ensuring alignment with user needs and market conditions. Development will adopt a Minimum Viable Product (MVP) strategy, prioritizing core features, robust synchronization and in-room text chat. Once stable, later sprints will introduce advanced features such as peer-to-peer video calling and interface refinements.

3.2 Requirements

3.2.1 Functional Requirements

For Fortheplot project, the key functional requirements are:

1. User and Account Management

- **User Authentication:** The system must allow users to register, log in, and manage their profiles.
- **User Roles:** The platform must support role-based permissions, distinguishing between a "Host" and a "Guest." Hosts will have specific controls over the viewing session.

2. Content and Player Features

- **Video Playback:** The platform must feature an HLS-based adaptive video player that can play movies and adjust quality based on network conditions.
- **Host Controls:** The host of a viewing session must have full control over video playback, including the ability to play, pause, and seek the video for all participants.

- **Content Catalog:** The system must provide a browsable movie catalog with search and filtering capabilities.
- **Content Limitation:** The project will only use public domain or self-uploaded demo films.

3. Real-Time Room System

- **Room Creation:** Users must be able to create private and public rooms.
- **Joining a Room:** Users should be able to join a room via an invitation link.
- **Real-time Synchronization:** The platform's core function is to achieve frame-accurate video synchronization across all participants.
- **Text Chat:** The system must include a persistent, low-latency, in-room text chat with message history.
- **Video Calling:** An optional peer-to-peer video calling feature must be integrated, supporting a maximum of four concurrent video streams.

4. Administration and Analytics

- **Admin Dashboard:** A basic administrative panel must be available for managing users and content.
- **Moderation:** The host must have moderation tools, such as the ability to kick disruptive users.
- **Content Management:** The admin dashboard should support CRUD (Create, Read, Update, Delete) operations for managing movie data.

3.2.2 Non-Functional Requirements

Non-functional requirements specify how the system should operate, including its performance, security, and usability. For the Fortheplot project, these include:

1. Performance and Scalability

- **Synchronization Reliability:** The synchronization engine must be resilient and mitigate playback drift by implementing adaptive streaming with buffer management and regular sync checkpoints.
- **User Capacity:** The platform is designed to support a maximum of 10 concurrent users per room.
- **Resolution:** Video resolution is capped at 1080p.

2. Reliability and Availability

- **Network Resilience:** The platform must be able to handle connection issues gracefully. It will use STUN servers to find a user's public IP address and port and will fall back to TURN servers as a relay when direct peer-to-peer connections are not possible due to restrictive firewalls.
- **Error Handling:** The system should have robust error detection and recovery mechanisms to handle issues like network failures or media decoding errors.

3. Security

- **Access Control:** The system must have secure communication built-in.
- **Form Validation:** Shared validation logic between the frontend and backend will ensure data integrity.
- **Auth:** User authentication will be handled using Spring Security to ensure that only authorized users have access to certain features.
- **Secure Servers:** STUN and TURN servers must be configured with robust authentication and security measures.

4. Usability and Maintainability

- **User Experience:** The user interface should be responsive and optimized
- **Code Quality:** The use of TypeScript provides end-to-end type safety, helping to catch breaking changes at compile-time instead of runtime.
- **Documentation:** The project's deliverables must include a complete set of documentation and a database schema.

3.2 Preliminary Investigation & Design Phase

Frontend: The interface will be developed with React.js and TypeScript. React's component-based model supports dynamic single-page applications, while TypeScript adds type safety. Automatic generation of types from the Spring Boot backend reduces runtime errors and improves development efficiency .

Backend: The backend will use Spring Boot, a mature Java framework known for building secure, high-performance applications. It simplifies deployment and provides native support for REST APIs and real-time WebSocket communication (GeeksforGeeks, 2023).

Database: A polyglot persistence approach will be adopted. PostgreSQL will store structured data such as user profiles and authentication records and movie metadata and content catalogs.

3.3 Front-End and Real-Time Systems

Video Synchronization: Playback will be managed with ReactPlayer and HLS.js. HLS segments video into small files, enabling adaptive quality based on network conditions to minimize buffering. Synchronization is maintained through WebSockets, which support bidirectional communication between client and server (GeeksforGeeks, 2023). The host controls the playback state, play, pause, or seek, and updates are broadcast to all participants, ensuring synchronized viewing.

Real-Time Text Chat: In-room chat will also use WebSockets to provide a persistent, low-latency communication channel. Each viewing session corresponds to its own WebSocket channel, limiting message delivery to participants in that room.

Multi-User Video Calling: Peer-to-peer video chat will be enabled through WebRTC and PeerJS. WebRTC supports direct browser-to-browser communication, reducing server load and bandwidth costs. PeerJS simplifies connection management and stream negotiation. To preserve performance, a maximum of four concurrent video streams will be supported per room.

3.4 Addressing Technical Complexities and Mitigation

Challenge 1: Synchronization Complexity

Problem: Achieving frame-accurate playback across users is difficult under varying network conditions, as small delays can accumulate into noticeable drift .

Mitigation: A hybrid method will combine client-side buffer management with server-side checkpoints. Clients will monitor bandwidth and buffer health using hls.js settings, while the host's playback position remains the master reference. A periodic "sync snap" will broadcast timecode updates to correct drift.

Challenge 2: WebRTC Connectivity

Problem: Peer-to-peer connections often fail behind restrictive firewalls or Symmetric NAT devices (Videosdk.live, 2025).

Mitigation: The platform will use STUN and TURN servers under the ICE protocol. STUN assists in establishing direct peer connections, while TURN acts as a fallback relay if direct routing fails. This guarantees reliable connections, though TURN introduces higher latency and bandwidth costs (Videosdk.live, 2025).

Challenge 3: Content Licensing

Problem: Licensing copyrighted films is legally complex and costly, especially for a

proof-of-concept.

Mitigation: The project will only use public domain, freely available, or self-uploaded demo films. This ensures the system remains academic, avoids legal risks, and demonstrates technical feasibility without commercial obligations.

CHAPTER 4: IMPLEMENTATION

4.0 System Analysis and Design

This chapter presents the implementation process of the Fortheplot Social Movie Streaming System, focusing on the analysis, design, and development phases. It outlines the system architecture through diagrams such as the Data Flow Diagram (DFD) and the Entity Relationship Diagram (ERD). Additionally, a proposed implementation timeline is provided to illustrate the structured development process.

The implementation was guided by the functional and non-functional requirements identified in earlier chapters, ensuring that all system modules, user authentication, real-time streaming, synchronized playback, chat integration, and administration, were cohesively designed and executed.

4.1 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) illustrates the logical flow of data within the Fortheplot system. It provides a visual representation of how information moves between users, system processes, and data stores.

4.1.1 Context Diagram (Level 0 DFD)

The Context Diagram (Level 0 DFD) represents the system as a single process and shows interactions with external entities such as Users, Hosts, and Administrators.

External Entities:

- **User:** Registers, logs in, browses movies, joins viewing rooms, and interacts via chat.

- **Host:** Creates viewing rooms, invites participants, and controls playback.

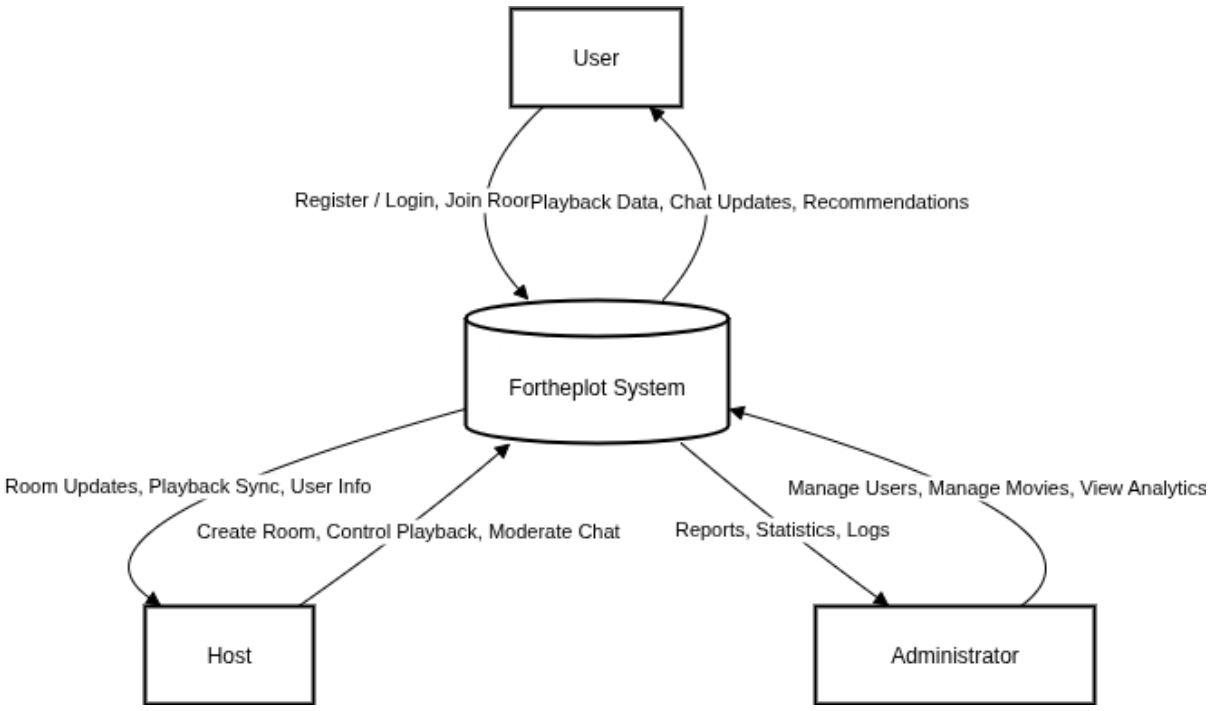
- **Administrator:** Manages users, movies, and system analytics.

Process Description:

- The system receives user inputs (login, room actions, movie selections).

- It communicates with the database for authentication, data retrieval, and synchronization.

- The system returns processed data (playback synchronization, chat updates, user lists) to the relevant actors.



4.1.2 Level 1 DFD

The Level 1 DFD breaks down the Fortheplot system into its main processes and data stores, providing more detail on internal data interactions.

Processes:

1. **User Management:** Handles registration, login, and profile updates.

2. **Movie Catalog Management:** Retrieves and displays available movies.

3. **Room Management:** Enables hosts to create and manage synchronized viewing rooms.

4. **Playback Synchronization:** Ensures all participants watch the movie in real time.

5. **Chat System:** Facilitates text-based communication during sessions.

6. **Admin Dashboard:** Allows monitoring of user activities and content management.

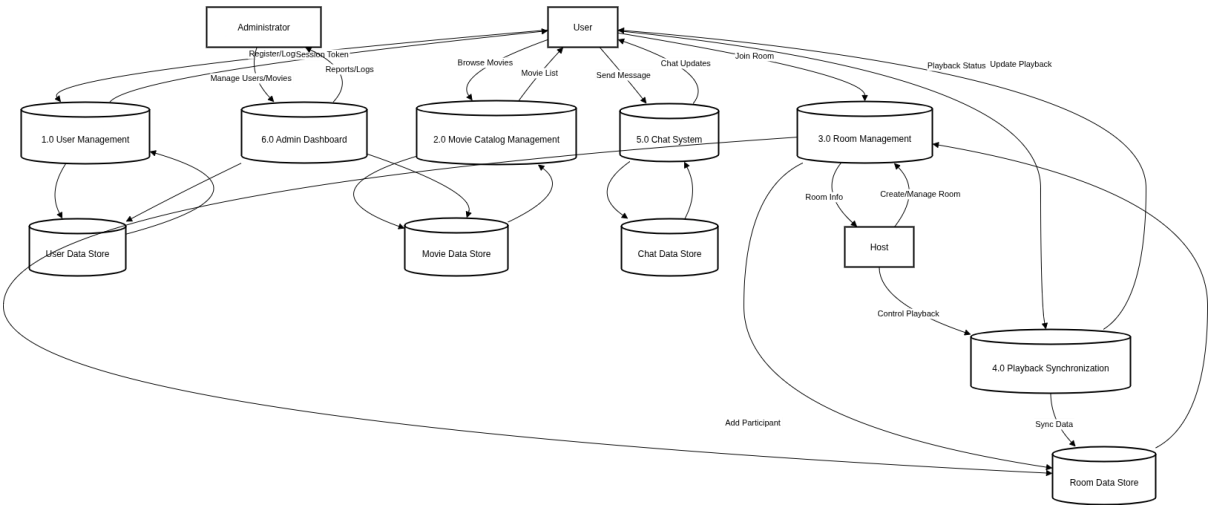
Data Stores:

- **User Data Store:** Contains user credentials and profiles.

- **Movie Data Store:** Stores movie metadata, links, and playback information.

- **Room Data Store:** Keeps records of active rooms and participants.

- **Chat Data Store:** Archives session messages for playback or moderation.



4.2 Entity Relationship (ER) Diagram

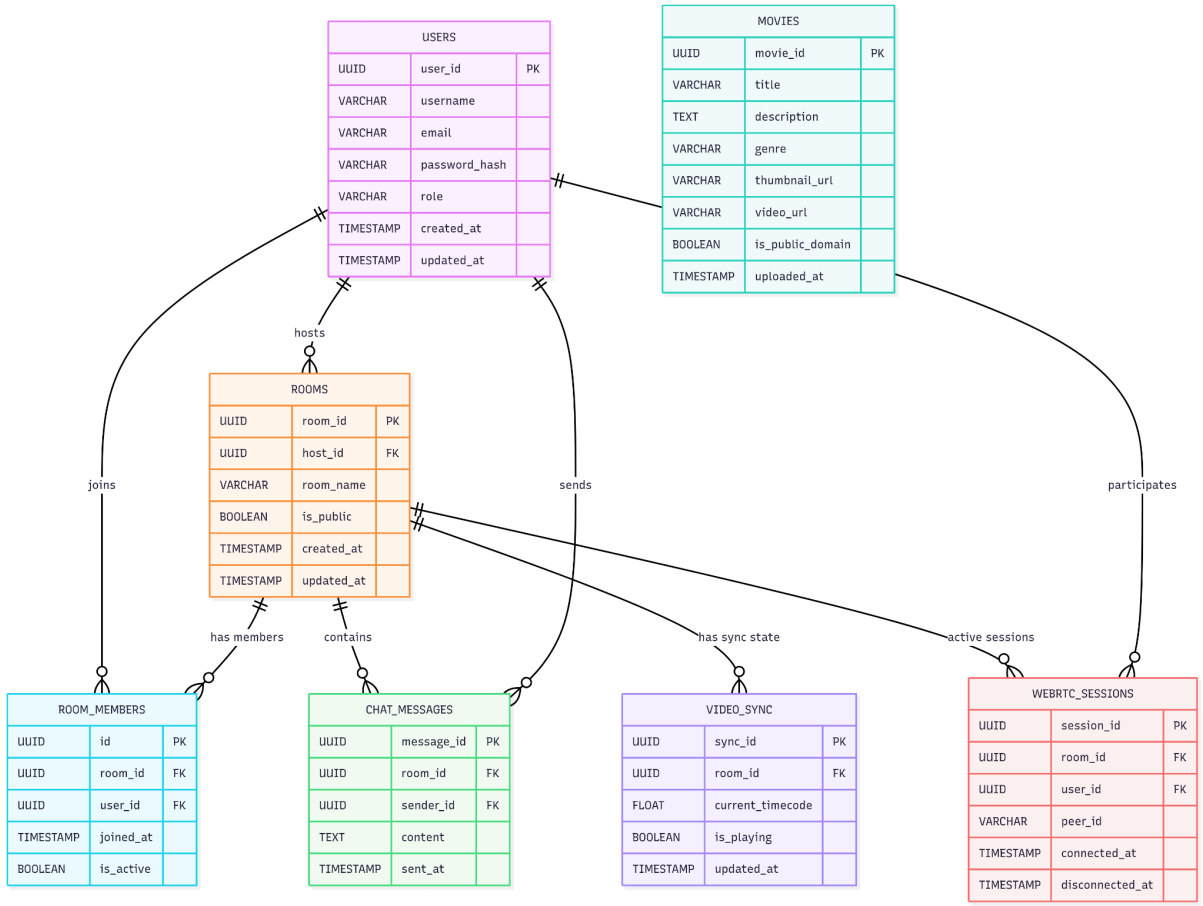
The Entity Relationship Diagram (ERD) models the database structure of the Fortheplot system. It identifies entities, their attributes, and the relationships among them.

Main Entities:

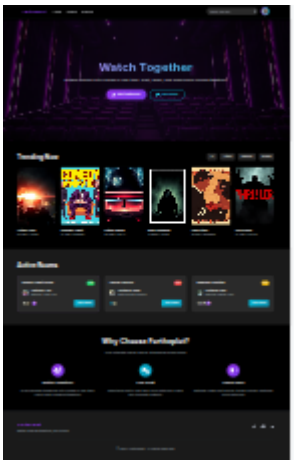
- User (UserID, Username, Email, PasswordHash, Role)
- Movie (MovieID, Title, Genre, Duration, URL)
- Room (RoomID, HostID, MovieID, CreatedAt, IsActive)
- ChatMessage (MessageID, RoomID, UserID, Content, Timestamp)
- Playback (PlaybackID, RoomID, CurrentTime, Status)

Relationships:

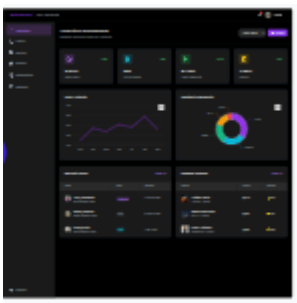
- A User can host multiple Rooms.
- A Room is associated with one Movie.
- A Room can have many ChatMessages.
- Each ChatMessage belongs to one User and one Room.



4.3 User Interface designs

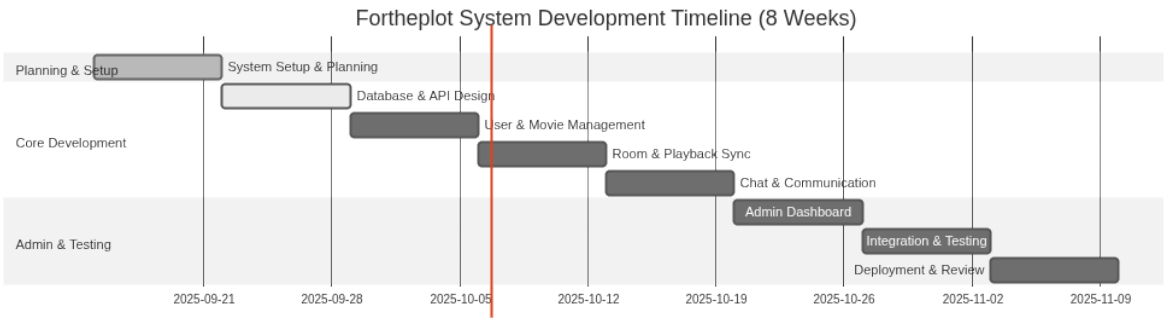






4.4 Proposed Timeline

Week	Activity	Description
Week 1	**System Setup & Planning**	Define project scope
Week 2	**Database & API Design**	Finalize project and proposal and create design diagrams and UI mockup
Week 3	**User and Movie Management**	Implement user registration/login, JWT token handling, and movie catalog browsing endpoints.
Week 4	**Room Creation & Synchronization**	Develop endpoints for room management, joining, playback sync logic, and host controls.
Week 5	**Chat & Communication Features**	Add real-time chat via SignalR and integrate message storage.
Week 6	**Admin Dashboard**	Implement admin panel for user, movie, and analytics management.
Week 7	**Integration & Testing**	Conduct unit and integration testing for backend and frontend; fix synchronization and chat bugs.
Week 8	**Final touches**	Conduct user testing and finalize documentation and report



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